



Modeling and performance analysis of a 5MW grid-connected photovoltaic power plant in the hot region of in Salah, Algeria

Modelação e análise de desempenho de uma central fotovoltaica de 5MW ligada à rede na região quente de Salah, Argélia

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ABSTRACT

This research evaluates the performance of a 5MW grid-connected photovoltaic (PV) power plant located in the In Salah region of southern Algeria, utilizing PVsyst simulation software. Commissioned in 2016, the plant spans across 10 hectares and employs polycrystalline silicon (Si-poly) modules for its solar energy generation. Detailed meteorological data specific to the In Salah region's hot desert climate,



including solar irradiation levels, temperature ranges, and other relevant parameters, were analyzed and incorporated into the simulation. The PVsyst modeling focused on accurately representing the PV system's configuration, energy production capabilities, potential losses, and overall performance ratio (PR). The annual simulation results revealed that the In Salah PV plant generated 7,603,339 kWh of energy over the course of a year. The PV array exhibited an efficiency of 12.67%, while the overall system achieved a performance ratio of 81.7%. A comprehensive loss diagram was constructed, identifying the major contributors to energy losses, such as temperature-related losses (7.6%), mismatch losses (2.1%), and inverter losses (5%). To assess the plant's performance in a broader context, the In Salah PV installation's PR was compared to other large-scale PV projects within Algeria and internationally. This comparative analysis highlighted that the In Salah plant's performance was average, influenced by factors such as the solar cell technology employed and the specific climatic conditions of the region. The study underscores the importance of utilizing simulation tools like PVsyst in the design and optimization phases of large-scale solar projects. Such simulations enable accurate evaluation of PV system performance under specific environmental conditions, while also identifying potential areas for efficiency improvement. This is particularly crucial for solar projects planned in hot regions, where the impact of high temperatures and other climatic factors on energy generation and system performance must be carefully considered.

Keywords: PV modeling, grid-connected PV, performance ration, energy yield analysis, power PV.

RESUMO

Esta investigação avalia o desempenho de uma central fotovoltaica (PV) de 5MW ligada à rede, localizada na região de In Salah, no sul da Argélia, utilizando o software de simulação PVsyst. Colocada em funcionamento em 2016, a central estende-se por 10 hectares e utiliza módulos de silício policristalino (Si-poly) para a produção de energia solar. Foram analisados e incorporados na simulação dados meteorológicos pormenorizados específicos do clima quente do deserto da região de In Salah, incluindo níveis de irradiação solar, intervalos de temperatura e outros parâmetros relevantes. Os resultados da simulação anual revelaram que a central fotovoltaica de In Salah produziu 7.603.339 kWh de energia ao longo de um ano. O conjunto fotovoltaico apresentou uma eficiência de 12,67%, enquanto o sistema global atingiu um rácio de desempenho de 81,7%. Foi construído um diagrama de perdas abrangente, identificando os principais factores que contribuíram para as perdas de energia, tais como as perdas relacionadas com a temperatura (7,6%), as perdas por desfaseamento (2,1%) e as perdas do inversor (5%). Para avaliar o desempenho da central num contexto mais amplo, a RP da instalação fotovoltaica de In Salah foi comparada com outros projectos fotovoltaicos de grande escala na Argélia e a nível internacional. Esta análise comparativa revelou que o desempenho da central de In Salah era médio, influenciado por factores como a tecnologia de células solares utilizada e as condições climáticas específicas da região. O estudo sublinha a importância da utilização de ferramentas de simulação como o PVsyst nas fases de conceção e otimização de projectos solares de grande escala. Estas simulações permitem uma avaliação exacta do desempenho do sistema fotovoltaico em condições ambientais específicas, ao mesmo tempo que identificam potenciais áreas de



melhoria da eficiência. Isto é particularmente crucial para projectos solares planeados em regiões quentes, onde o impacto das altas temperaturas e de outros factores climáticos na produção de energia e no desempenho do sistema deve ser cuidadosamente considerado.

Palavras-chave: modelação FV, FV ligado à rede, razão de desempenho, análise de rendimento energético, potência FV.

1 INTRODUCTION

Renewable energy is a major global challenge today as energy demand soars with population and industrial growth. By 2020, global renewable capacity including 714 GW of solar power exceeded expectations, breaking records. Over 257 GW of new renewable capacity was added in 2020 despite the economic slowdown, 59% higher growth than 2019 [Mohammadi *et al.* 2018]. To meet rapidly rising energy needs, photovoltaic solar power is increasingly being used to generate electricity as a clean, renewable alternative to polluting fossil fuels. Many countries are seeking to harness their solar resources and develop photovoltaic technologies to produce electricity [Abada; Bouharkat 2018; Bouznit *et al.* 2020a; Bouzidi *et al.* 2022]

This is due to the increasing energy consumption, which is expected to double by 2060, and as a result of the increasing demand for energy and global climate change, there are new policies calling for the deployment of renewable energy [Chahar; Fenni 2022a]. Therefore, the Paris Climate Agreement holds participating countries and companies responsible for achieving the sustainability goals they have set for themselves by 2030 [Fenni 2022b; Ihaddadene *et al.* 2022; Bouzidi *et al.* 2024].

Algeria is one of the developing countries and is a country located in North Africa within the Arab Maghreb. It is the largest country in the Mediterranean Sea and the largest in area in Africa. It is bordered to the northeast by Tunisia, to the east by Libya, to the south by Mali and Niger, to the southwest by Mauritania and Western Sahara, and to the west by Morocco. Its area is approximately 2,381,741 square kilometers, and its population is 43.9 million [Abada; Bouharkat 2018]. Algeria's climate is of the Mediterranean type along the entire northern edge, which covers the coast and the Atlas Hills (hot dry summer, wet and cold winter), semi-arid on the high plateaus in the center of the country, and desert once you cross



the Saharan Atlas Mountains. Temperature differences can be significant in a single day, as is the case in the desert, where temperatures can fluctuate from one end to the other within a few hours (more than 40°C during the day – less than 5°C at night). The Sahara covers 85% of Algerian territory (2,000 km from west to east and 1,500 km from north to south) [75 2023; Bouznit *et al.* 2020b].

Like other developing nations, Algeria developed a national renewable energy program aiming to meet 40% of its electricity needs from renewables, mainly solar photovoltaics, by 2030 due to its abundant sunshine across 90% of its 2.38 million km² land area averaging 3,000 hours and 5kW/m² annually. The Sahara Desert, from which the Arabic word meaning "plateau land" originates, covers much of Algeria and experiences extremely hot and dry conditions, reaching 50°C in summer days and below 5°C in winter highlands [Dincer; Ozer 2023; Zgham *et al.* 2024]

Regarding photovoltaic power plants, their modeling, and behavior analysis, many researchers worldwide have addressed these topics in their research papers, such as [Bouziane *et al.* 2022; Dincer and Ozer 2023; Faiz *et al.* 2021; Muhammad *et al.* 2023; Shiva Kumar; Sudhakar 2015].

This paper aims to evaluate the performance of a 5MW grid-connected photovoltaic power plant located in the hot region of In Salah, southern Algeria, utilizing the widely-used PVsyst simulation software. The well-known PVsyst tool was selected to comprehensively study and analyze the performance characteristics of this 5MWp solar power facility in the In Salah area.

2 CASE STUDY REGION AND EMPLOYED DATA

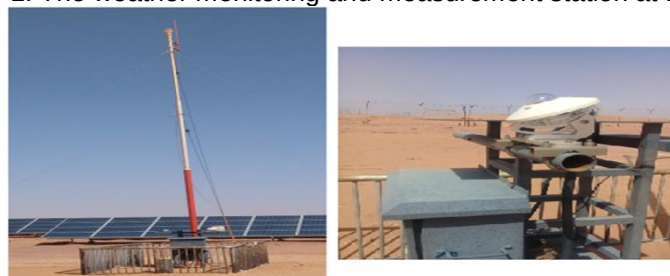
2.1 THE SOLAR POWER STATION IN IN SALAH PROVINCE, SOUTHERN ALGERIA

The solar power station, known as SKTM, is strategically located in In Salah province, southern Algeria, at the coordinates 2°30'17" East and 27°10'54" North. This strategic geographical position makes it an ideal site for the development of renewable energy projects, particularly solar power generation. The In Salah region is part of the vast Sahara Desert, endowed with rich natural resources like oil and natural gas, further enhancing its suitability for harnessing solar energy [Fezzani *et al.* 2023]. Commissioned on 11 February 2016, the SKTM solar power

station boasts an impressive production capacity of (5 MW) , spanning an area of approximately 10 hectares. This large-scale project is a crucial component of Algeria's National Program for Renewable Energies and Energy Efficiency, aimed at strengthening the electrical grid, diversifying and developing electricity production sources, preserving non-renewable resources, reducing emissions, and protecting the environment. The station's infrastructure comprises five subfields, five electrical transformers, a main transformer, and numerous connection and parallel boxes, totaling an impressive 20,460 solar panels. This extensive solar array not only contributes significantly to meeting the region's energy needs but also plays a vital role in achieving environmental sustainability objectives. By harnessing the abundant solar resources of the Sahara Desert, the SKTM solar power station in In Salah province exemplifies Algeria's commitment to transitioning towards a more sustainable energy future, reducing reliance on fossil fuels, and mitigating the impacts of climate change.

The station is equipped with an integrated weather monitoring system featuring sensors, communication, data aggregation, and power supply units for efficient operation monitoring, reporting, and control. It utilizes a modular, expandable design to measure weather data, process signals, perform calculations, and transfer information to the main station. Figure 1.

Figure 1. The weather monitoring and measurement station at In Salah



Source: The authors.

As for the type of photovoltaic panel used at the In Salah station is (YL245P-29b) it is known for its exceptional performance and durability. Featuring high-efficiency polycrystalline solar cells, it maximizes energy conversion from sunlight. Its high transmission and textured glass enhance light absorption, ensuring optimal output even in varied weather conditions. With a module series efficiency of up to 16.5% in mono-crystalline cells and 15.4% in multi-crystalline cells, it delivers



reliable power generation. The panel boasts a power tolerance of $\pm 3\%$, providing consistent performance over its lifespan. Encased in a robust, corrosion-resistant aluminum frame, it offers long-term durability, backed by a 25-year limited power warranty. The YL245P-29b is part of the high-quality YGE Series modules and is UL certified to 1000V, ensuring safety and reliability for all applications.

2.2 WEATHER CONDITIONS AND DATA COLLECTION

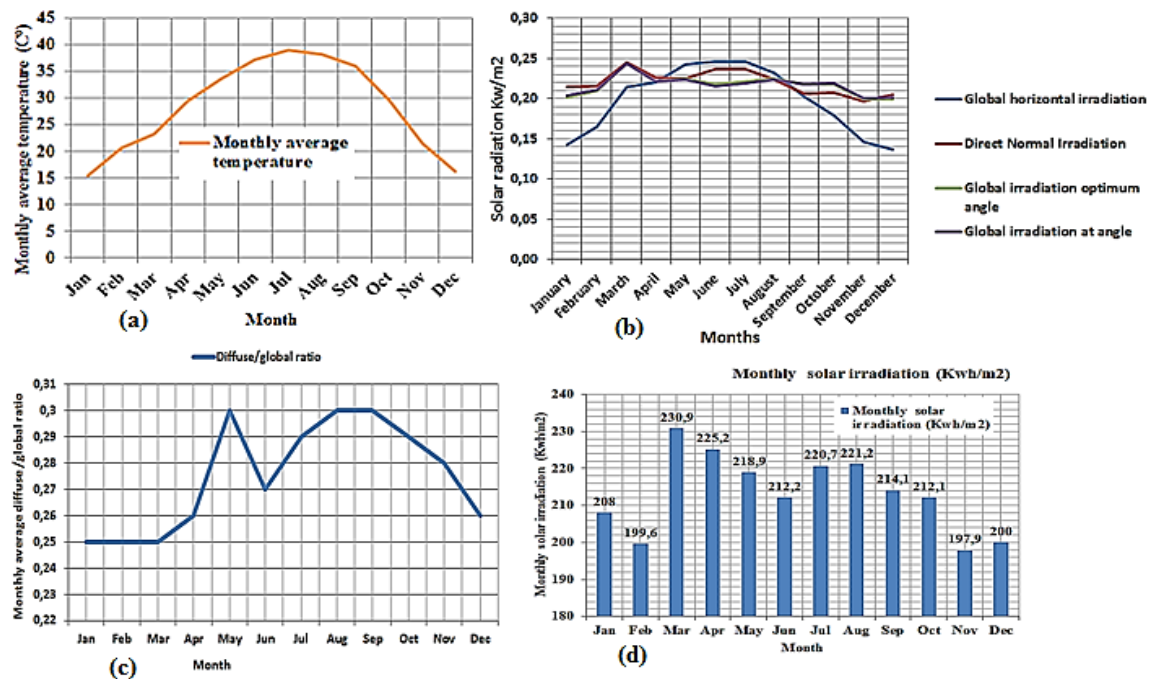
The climate of In Salah is characterized by low rainfall; with an annual average not exceeding 25 mm. summer temperatures soar to record highs during the day and drop at night. Winters are warm, and strong sandstorms occur in spring. The hot Sirocco winds in summer can sometimes raise temperatures to over 50°C , which has previously led to the deaths of many infants and elderly people [Dabou *et al.* 2016; Fezzani *et al.* 2023], During the year, the monthly average temperature peaked at a maximum of 39°C in July and a minimum of 15.5°C in January. There was a gradual increase from January to July, followed by a decline from August to December Figure (2.a)

And the global horizontal irradiation averaged a maximum of (243.28 kWh/m²) in May and a minimum of (142.16 kWh/m²) in January. For direct normal irradiation, March had the highest monthly level at (244.52 kWh/m²), while September had the lowest at 205.79 kWh/m². Regarding global irradiation at the optimum angle, March was the most favorable month with (243.49 kWh/m²), whereas January was the lowest at 201.77 kWh/m². Finally, for a fixed 28° angle, global irradiation peaked at (243.83 kWh/m²) in March and dipped to 203.85 kWh/m² in January. Overall, these data show that insolation levels reached their maximum in spring and summer, and their minimum in winter Figure (2.b).

The monthly average ratio of diffuse to global irradiation was highest at 0.3 in October and lowest at 0.25 in January. The ratio gradually increased from January to October, then slightly decreased in November and December Figure (2.c). For monthly in-plane irradiation on flat solar panels, it peaked at 230.9 kWh/m² in March and troughed at 197.9 kWh/m² in November Figure (2.d). Overall, these data provide valuable information for sizing and optimizing the photovoltaic installation to maximize annual solar electricity generation.



Figure 2. (a): Monthly average temperature, (b): Global horizontal irradiation, Direct Normal Irradiation, Global irradiation optimum angle Global irradiation at angle, kWh/m2, (c): Monthly average diffuse to global ratio, (d): Monthly in-plane irradiation for fixed-angle



Source: The authors.

3 PV SYSTEM PERFORMANCE PARAMETERS

The International Energy Agency (IEA) has developed performance parameters [Faiz *et al.* 2021; Ihaddadene *et al.* 2023] for analyzing the performance of grid-connected photovoltaic systems. These parameters would provide a comprehensive pre-feasibility analysis for a solar photovoltaic power plant. Among the most important of these parameters, we will address:

3.1 ARRAY YIELD(YA)

Array Yield (Ya) refers to the DC energy produced by the photovoltaic array, calculated as the ratio of the total DC energy generated to the nominal array power. The daily array yield can be expressed by the equation [Faiz *et al.* 2021; Rabczak; Proszak-Miąsik 2020a]:

$$Y_a = \frac{E_{DC}}{P_{PV(STC)}} \quad (1)$$

Where:



$$E_{DC} = V_{DC} * I_{DC} * t \quad (2)$$

3.2 REFERENCE YIELD

The Reference Yield (Y_r) represents the total energy that would be produced by the system operating at its nominal rated efficiencies. This nominal system efficiency is pre-defined by the array manufacturers under Standard Test Conditions (STC) and listed on the nameplate details. Y_r is mathematically expressed as the ratio of the total horizontal irradiance to the global irradiance at STC of (1000 W/m^2). Its units are $\text{kWh/m}^2/\text{day}$ [Kumar *et al.* 2017]. The reference yield (Y_r) is the ratio between the total solar radiation incident on the PV array surface and the reference irradiance of 1000 W/m^2 :

$$Y_r = \frac{H_t}{G_0} \quad (3)$$

3.3 FINAL SYSTEM YIELD

The final system yield, denoted as Y_f , is the amount of energy injected into the grid annually, monthly or daily. It is defined as the ratio of the AC energy output of the photovoltaic system (inverter output) to the peak installed power of the PV array under standard test conditions (STC). The daily and monthly final yields are calculated using specific formulas that relate the total AC energy generated to the installed nominal power [Dalal *et al.* 2020; Ihaddadene *et al.* 2023]:

$$Y_f = \frac{E_{AC}}{P_{PV(STC)}} \quad (4)$$

3.4 NORMALIZED PERFORMANCE RATIO (PR)

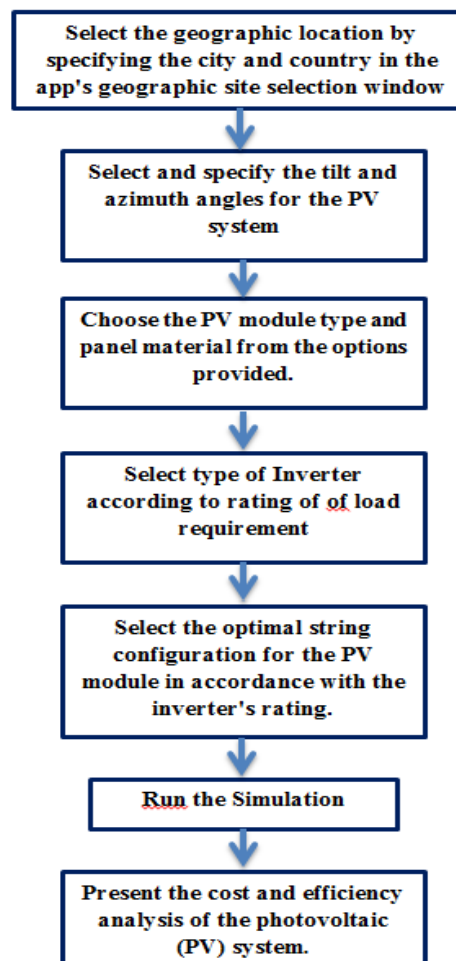
The normalized performance ratio (PR) quantifies the overall effect of losses on the energy production of a photovoltaic (PV) system's arrays. PR values indicate how closely the actual performance of a PV system approaches its ideal performance under operating conditions. It is a dimensionless quantity defined as the ratio of final to reference energy yields [Dalal *et al.* 2020; Kumar *et al.* 2017; Rabczak; Proszak-Miąsik 2020b]:

$$PR = \frac{Y_f}{Y_r} = \frac{E_{grid}}{GI_{inc}} \quad (5)$$

4 PVSYST SIMULATION MODELING

PVsyst is standard simulation software. It is widely used by engineers globally for its rapid results, user-friendliness, and comprehensive analysis of parameters influencing system efficiency. It can perform periodic estimations, generate reports, and provides high accuracy close to real values. It also features color-coding for displaying errors and warnings [Kumar *et al.* 2017; Zgham *et al.* 2024]. The simulation models a grid-connected polycrystalline silicon (Si-poly) PV system sized per project specifications using the PVsyst tool, consisting of PV modules, inverters, and grid interface. The design methodology steps involved in the simulation are illustrated as a flowchart in Figure 3. The step-by-step procedure for simulating a PV system design in PVsyst includes.

Figure 3. Step-by-step analysis process for PVsyst



Source: The authors.

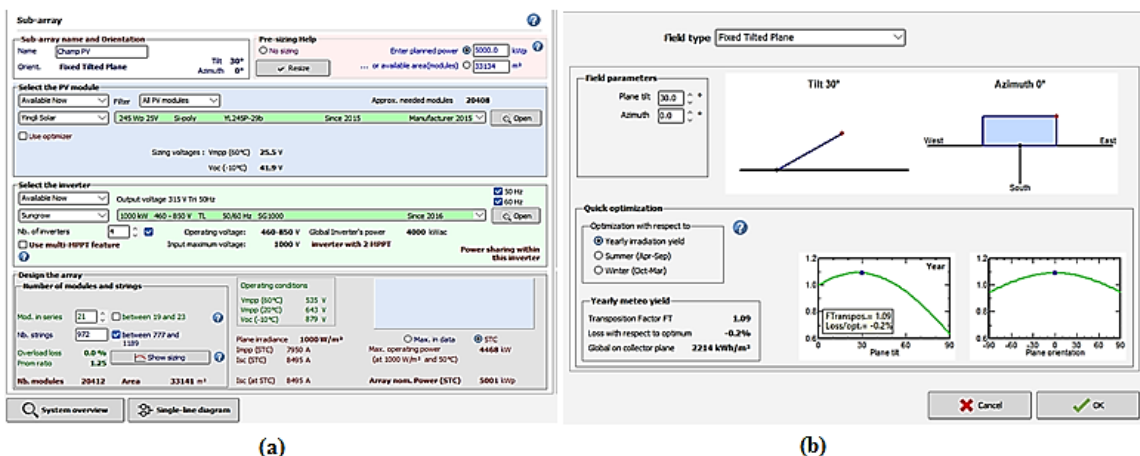
5 RESULT AND DISCUSSION

This section analyzes the simulation results of a proposed (5MW) Si-poly photovoltaic system using PVsyst. The analysis focuses on key metrics including energy production, specific production, performance ratio, and losses to evaluate the system's performance.

The "Orientation" section allows specifying the orientation of the PV system components. The "Tracking" option enables automatic sun-tracking for optimized energy production. Without tracking, you can manually set the tilt angle (modules' inclination from horizontal) and azimuth angle (compass direction modules face). These settings significantly impact energy yield and should be carefully chosen based on latitude and local solar radiation patterns. For the studied site, the optimal tilt angle is 27° , and the azimuth angle is 0° (true south), as shown in Figure (4.a). After orientation, the simulation is conducted by selecting the field type, parameters, and optimization criteria as shown in Figure (4.b).

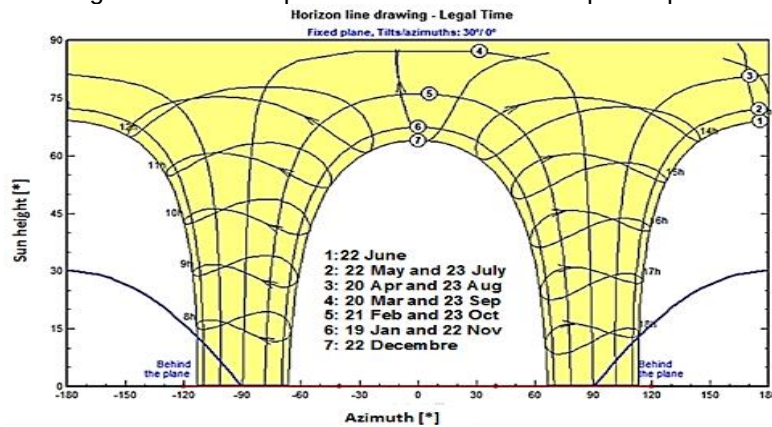
Figure 5 shows the sun path diagram for In Salah, southern Algeria, highlighting the significant variation in daylight hours throughout the year. The shortest day on December 21st has around 10.5 hours of daylight, while the longest day in June has approximately 13.5 hours. It also illustrates the sunrise and sunset times, as well as the azimuth angles (compass direction) of the sun throughout the day, ranging from east (90 degrees) to west (270 degrees), with varying angles across seasons.

Figure 4. (a): Layout and key configurations of the project page, (b):Page simulating Solar panel orientation



Source: The authors.

Figure 5. The sun path at the In Salah solar power plant



Source: The authors.

5.1 THE MAIN SIMULATION OUTPUTS AND ASSOCIATED BALANCES WERE ANALYZED

From the main simulation results, three key parameters were evaluated. Firstly, the total energy produced annually by the 5 MWp polycrystalline silicon photovoltaic system, referred to as produced energy, was 7,603,339 kWh/year. Secondly, the annual average performance ratio (PR) was 81.7%.

Table 1 shows irradiance values (global horizontal, on collector plane with/without optical corrections), ambient temperature, DC energy from PV array, AC energy to grid, and PV array/system efficiencies. Monthly and yearly values were calculated, with yearly values being averages for temperature/efficiency and sums for irradiance/energy.

Table 1. Balance and main results of the system obtained from the PVSyst (5MW_p)

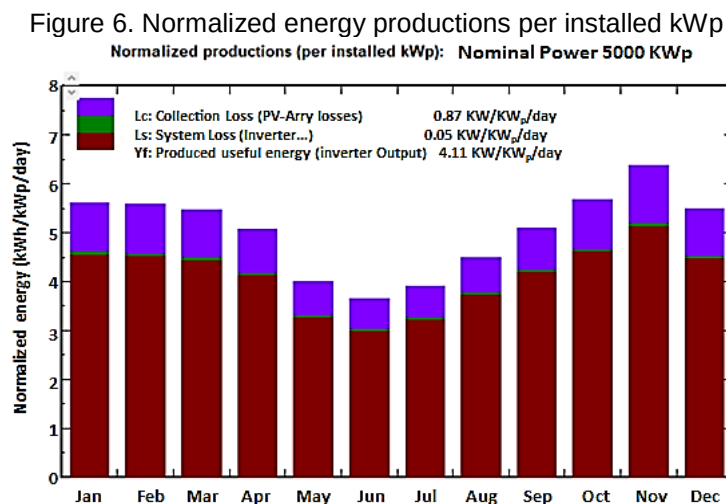
	GlibHor KW/m2	DiffHor KW/m2	T_Amb C°	GlobInc KW/m2	GLobEff KW/m2	E_Array KWh	E_Grid KWh	Eff ArrayR (%)	Eff SysR (%)
Jan	158,3	85,07	26,37	173,8	171	717028	708277	14,34056	14,16554
Feb	152,4	83,8	26,97	156,7	153,9	642555	634740	12,8511	12,6948
Mar	180,6	89,27	27,3	169,8	165,8	698377	690037	13,96754	13,80074
Apr	181,4	76,84	26,74	152,1	147,2	627838	620318	12,55676	12,40636
May	162,1	81,56	26,39	124	118,1	514674	508337	10,29348	10,16674
Jun	149,7	71,26	24,56	109,4	103,5	457276	451557	9,14552	9,03114
Jul	163,7	74,32	24,2	120,9	114,8	507278	501061	10,14556	10,02122
Aug	173,2	80,22	23,65	139,5	134,1	588835	581819	11,7767	11,63638
Sept	169,9	71,6	24	153,1	148,7	639701	631943	12,79402	12,63886
Octo	176,3	73,67	25,41	176,2	172,3	725167	716531	14,50334	14,33062
Nou	173,5	66,36	26,05	191,5	188,2	780682	771683	15,61364	15,43366
Dec	152,5	83,78	26,6	170,4	167,5	703928	695533	14,07856	13,91066
Year	1993,6	937,75	25,68	1837,4	1785,1	7603339	7511836	12,67223167	12,5197267

Source: The authors.

For the study location annually: global horizontal irradiance 1996 kWh/m², incident on collector without/with optical losses 1837,4/1785,1 kWh/m². DC energy from PV array 7603339 kWh, AC to grid 7511836 kWh PV array efficiency 12.67%, system efficiency 12.52%.

5.2 NORMALIZED PRODUCTIONS

The simulation study evaluated various energy productions, including collection losses, system losses, and produced useful energy, per installed kWp per day, as illustrated in Figure 6. Lc represents the collection losses or the PV array capture losses, which is 0,43kWh/kWp/day, Ls is the system loss, amounting to 0,11kWh/kWp/day, and Yf is the produced useful energy, quantified as 4,62kWh/kWp/day. The recorded values show that the LC value is 0,43kWh/kWp/day the LS value is 0,11kWh/kWp/day, and Yf, the produced useful energy, is given as 4,62kWh/kWp/day



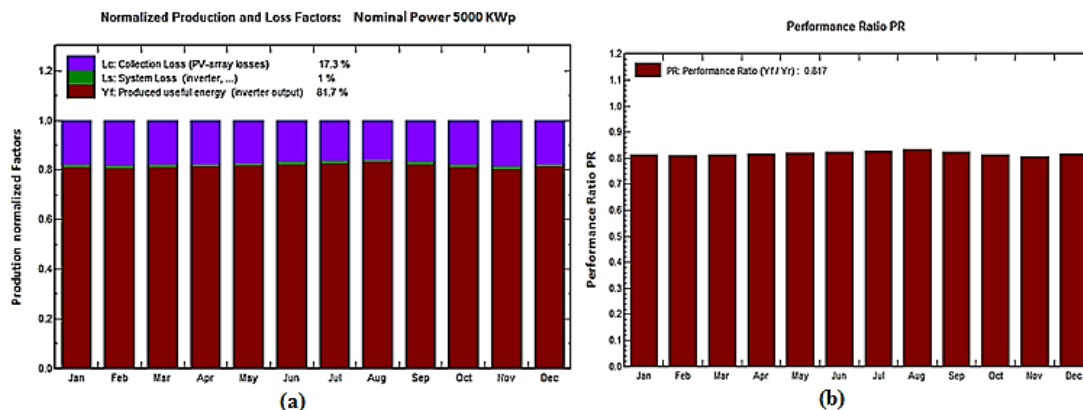
Source: The authors.

5.3 THE GRID-INJECTED ENERGY AND PERFORMANCE RATIO WERE ANALYZED

The (5MWp) Si-poly photovoltaic plant converts the DC energy produced by the PV array into AC before injecting it into the grid, resulting in some energy loss. It injects 7511836 kWh of energy into the grid annually. The average monthly and daily specific electricity injected is 125,19kWh/kWp and 4,17kWh/kWp respectively. March saw the highest energy injection of 690037 kWh, while July had the lowest at 451557 kWh. The detailed energy injection data is tabulated in

Table 2 and Figure (7.a) The simulated (5MWp) Si-poly system has an annual average Performance Ratio (PR) of 81.7%, with slight monthly variations shown in Figure (7.b).

Figure 7 (a): Normalized productions and Loss Factors,(b): Performance Ratio



Source: The authors.

Table 2. Energy injected to grid & (PR) ratio: PV (5000KWp)

	E_Grid KWh	Em_Grid (KWh/KWp)	Ed_Grid (KWh/KWp)	PR Ratio (%)
Jan	708277	141,65	4,72	81,5
Feb	634740	127	4,23	81
March	690037	138	4,6	81,3
Apr	620318	124	4,13	81,5
May	508337	101,67	3,4	82
Jun	451557	90,3	3,01	82,5
Jul	501061	100,2	3,34	82,8
Aug	581819	116,36	3,87	83,4
Sept	631943	126,4	4,2	82,5
Octo	716531	143,3	4,8	81,3
Noune	771683	154,3	5,14	80,6
Dece	695533	139,1	4,6	81,6
Year	7511836	125,19	4,17	81,7

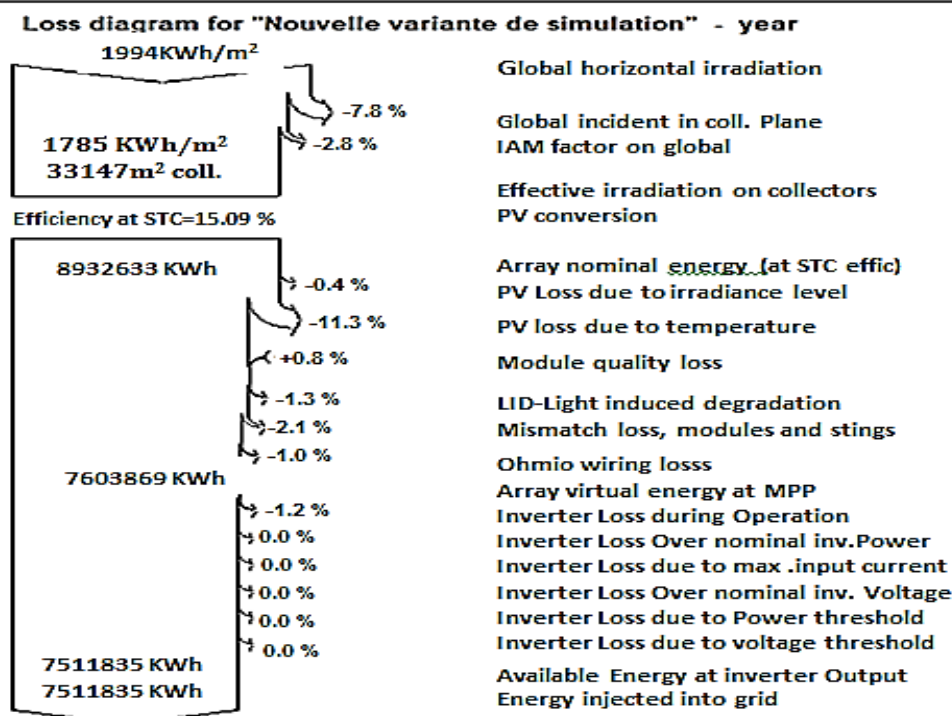
Source: The authors.

5.4 ARROW LOSS DIAGRAM

The power loss diagram, derived from simulated studies, aids in analyzing the various losses encountered during the installation of a photovoltaic (PV) plant or the constraints that need to be considered. Figure 8 shows the arrow loss diagram, illustrating the different system losses. The global irradiance on the horizontal plane is 1930 kWh/m², but the effective irradiance on the collector is 1857 kWh/m². This results in a 3.8% energy loss due to irradiance levels. When this effective irradiance falls on the surface of a PV module or array, electrical

energy is produced. After the PV conversion, the array's nominal energy at standard testing conditions (STC) is 8448 kWh, with an efficiency of 15.3%. The annual array virtual energy at maximum power point (MPP) is 7804 kWh. At this stage, various losses occur: 7.6% due to temperature, 2% due to light-induced degradation, 2.1% due to module array mismatch, and 1% due to Ohmic wiring losses. The annual available energy at the inverter output is 7240 kWh, which is then injected into the grid. Two possible losses at this point include a 3.5% inverter loss during operation and a 1.5% loss over nominal inverter power.

Figure 8. Power loss diagram of the 5 MWp grid-tied photovoltaic system



Source: The authors.

5.5 COMPARISON OF THE SOLAR POWER PLANT WITH OTHER PLANTS AROUND THE WORLD

The performance ratio (PR) of the In Salah, Algeria solar photovoltaic plant (81.7%) appears to be average compared to other PV plants mentioned. Some plants in Algeria like Sidi-bel-Abbés (88.3%) and Saida (85.52%) outperform it, while others in Adrar (73.68% & 74.36%) and M'Sila (71.59%) underperform. Outside Algeria, a Malaysian plant (85.4%) has a higher PR, while plants in Abu Dhabi (70% & 80%) have lower PR, and a Turkish plant (81.15%) is similar to In Salah. The variation in PR can be attributed to factors like solar cell technology,



panel tilt angle, and climatic conditions. In summary, the In Salah plant's performance is moderate relative to the other plants listed in the Table 3:

Table 3. Performance Ratio (PR) of solar photovoltaic plants across different locations

Place	PV Capacity (MWp)	PV Technology	Final Yield (Yf)	PR(%)	References
In Salah, Algeria	5	Si-Poly	4,11	81,7	Present Study
Sidi-bel-Abbés, Algeria	12	Si-Poly	4,15	83,01	[Fezzani <i>et al.</i> 2023]
Adrar, Algeria	6	Si-Poly	5,15	73,68	[Dahbi <i>et al.</i> 2021]
Adrar, Algeria	20	Si-Poly	4,98	74,36	[Bentouba <i>et al.</i> 2021]
Saida, Algeria	30	Si-Poly	4,9	85,52	[Ihaddadene <i>et al.</i> 2022]
El Bayadh, Algeria	23,9	Si-Poly	4,95	82,02	[Dahmoun <i>et al.</i> 2021]
Sidi-bel-Abbés, Algeria	60	Si-Poly	4,68	88,3	[Fezzani <i>et al.</i> 2023]
Malaysia (rooftop PV)	0.2325	Si- mono	---	85,4	[Saleheen <i>et al.</i> 2021]
M'Sila, Algeria	20	Si-Poly	3.99/5.897	71,59	[Kichene <i>et al.</i> 2023]
Turkey	2,13	Si-Poly	4,53	81,15	[Cubukcu and Gumus 2020]
Abu Dhabi (rooftop PV)	0,2157	Si- mono	3,63	70	[Ali and Emziane 2013]
	0,1144	Si-Poly	4,16	80	
	0,994	Si- mono	3,94	---	

Source: The authors.

6 CONCLUSIONS

This research presents a comprehensive evaluation of a 5MWp grid-connected photovoltaic solar power plant situated in the In Salah region of Algeria, utilizing the PVsyst simulation software. Through rigorous analysis, key findings are highlighted: the plant generates an annual energy output of 7,603,339 kWh with a performance ratio of 81.7%, various energy losses within the system are identified and quantified via a detailed loss diagram analysis, and the plant's performance is comparatively assessed against other photovoltaic installations in Algeria and globally, exhibiting an average performance.

Crucially, the study underscores how factors such as solar cell technology, panel tilt angle, and climatic conditions significantly influence the performance ratio.

By comprehensively evaluating solar power plant performance and pinpointing critical factors affecting efficiency, this research provides invaluable insights for optimizing the design and operation of photovoltaic systems. These insights hold the potential to maximize energy yield and enhance the long-term economic viability of solar installations, contributing to the advancement of renewable energy solutions and sustainable practices.



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